

Questions 1-10

- Before the mid-1860's, the impact of the railroads in the United States was limited, in the sense that the tracks ended at the Missouri River, approximately the centers of the country. At that point the trains turned their freight, mail, and passengers over to steamboats, wagons, and stagecoaches. This meant that wagon freighting, stagecoaching and steamboating did not come to an end when the first train appeared; rather they became supplements or feeders. Each new "end-of-track" became a center for animaldrawn or waterborne transportation. The major effect of the railroad was to shorten the distance that had to be covered by the older, slower, and more costly means. Wagon freighters continued operating throughout the 1870's and 1880's and into the 1890's, although over constantly shrinking routes, and coaches and wagons continued to crisscross the West wherever the rails had not yet been laid.

- The beginning of a major change was foreshadowed in the later 1860's, when the Union Pacific Railroad at last began to build westward from the Central Plains city of Omaha to meet the Central Pacific Railroad advancing eastward from California through the formidable barriers of the Sierra Nevada. Although President Abraham Lincoln signed the original Pacific Railroad bill in 1862 and a revised, financially much more generous version in 1864, little construction was completed until 1865 on the Central Pacific and 1866 on the Union Pacific. The primary reason was skepticism that a railroad built through so challenging and thinly settled a stretch of desert, mountain, and semiarid plain could pay a profit. In the words of an economist, this was a case of "premature enterprise", where not only the cost of construction but also the very high risk deterred private investment. In discussing the Pacific Railroad bill, the chair of the congressional committee bluntly stated that without government subsidy no one would undertake so unpromising a venture; yet it was a national necessity to link East and West together.

1. The author refers to the impact of railroads before the late 1860's as "limited" because
 - (A) the tracks did not take the direct route from one city to the next
 - (B) passenger and freight had to transfer to other modes of transportation to reach western destinations
 - (C) passengers preferred stagecoaches
 - (D) railroad travel was quite expensive
2. The word "they" in line 5 refers to
 - (A) tracks
 - (B) trains
 - (C) freight, mail, and passengers
 - (D) steamboats, wagons, and stagecoaches
3. The word "supplements" in line 6 is closest in meaning to
 - (A) extensions
 - (B) reformers
 - (C) dependents
 - (D) influences
4. What can be inferred about coaches and wagon freighters as the railroads expanded?
 - (A) They developed competing routes.
 - (B) Their drivers refused to work for the railroads.
 - (C) They began to specialize in transporting goods.
 - (D) They were not used as much as before.

5. The word "crisscross" in line 11 is closest in meaning to
(A) lead the way (B) separate
(C) move back and forth (D) uncover
6. Why does the author mention the Sierra Nevada in line 15?
(A) To argue that a more direct route to the West could have been taken
(B) To identify a historically significant mountain range in the West
(C) To point out the location of a serious train accident
(D) To give an example of an obstacle faced by the Central Pacific
7. The word "skepticism" in line 18 is closest in meaning to
(A) doubt (B) amazement (C) urgency (D) determination
8. The Pacific railroads were considered a "premature enterprise" (line 21) because
(A) the technology of railroad cars was not fully developed
(B) there was not enough wood and steel for the tracks
(C) the cost and risks discouraged private investment
(D) there were insufficient numbers of trained people to operate them
9. The word "subsidy" in line 23 is closest in meaning to
(A) persuasion (B) financing (C) explanation (D) penalty
10. Where in the passage does the author give example of geographical challenges to railroad construction?
(A) Lines 4-6 (B) Lines 8-11 (C) Lines 18-20 (D) Lines 22-25

Questions 11-22

Humanity's primal efforts to systematize the concepts of size, shapes, and number are usually regarded as the earliest mathematics. However, the concept of number and the counting process developed so long before the time of recorded history (there is
Line archaeological evidence that counting was employed by humans as far back as 50,000
(5) years ago) that the manner of this development is largely conjectural. Imagining how it probably came about is not difficult. The argument that humans, even in prehistoric times, had some number sense, at least to the extent of recognizing the concepts of more and less when some objects were added to or taken away from a small group, seems fair, for studies have shown that some animal possess such a sense.

(10) With the gradual evolution of society, simple counting became imperative. A tribe had to know how many members it had and how many enemies, and shepherd needed to know if the flock of sheep was decreasing in size. Probably the earliest way of keeping a count was by some simple tally method, employing the principle of one-to-one correspondence. In keeping a count of sheep, for example, one finger per sheep could
(15) be turned under. Counts could also be maintained by making scratches in the dirt or on a stone, by cutting notches in a piece of wood, or by tying knots in a string.

Then, perhaps later, an assortment of vocal sounds was developed as a word tally against the number of objects in a small group. And still later, with the refinement of writing, a set of signs was devised to stand for these numbers. Such an imagined
(20) development is supported by reports of anthropologists in their studies of present-day societies that are thought to be similar to those of early humans.

11. What does the passage mainly discuss?

- (A) The efforts of early humans to care for herds of animals
- (B) The development of writing
- (C) The beginnings of mathematics
- (D) Similarities in number sense between humans and animals

12. The word "conjectural" in line 5 is closest in meaning to

- (A) complex
- (B) based on guessing
- (C) unbelievable
- (D) supported by careful research

13. Why does the author mention animals in line 9?

- (A) To support a theory about the behavior of early humans
- (B) To identify activities that are distinctly human
- (C) To illustrate the limits of a historical record of human development
- (D) To establish that early human kept domesticated animals

14. The word "it" in line 11 refers to

- (A) evolution
- (B) counting
- (C) tribe
- (D) shepherd

15. What is the basic principle of the tally method described in the second paragraph?

- (A) The count is recorded permanently.
- (B) Calculations provide the total count.
- (C) Large quantities are represented by symbols.
- (D) Each marker represents a singly object.

16. The word "employing" in line 13 is closest in meaning to

- (A) using
- (B) paying
- (C) focusing
- (D) hiring

17. Which of the following is NOT mentioned as an early methods of counting?
(A) Cutting notches (B) Bending fingers
(C) Piling stones (D) Tying knots
18. The word "maintained" in line 15 is closest in meaning to
(A) justified (B) asserted (C) located (D) kept
19. The word "assortment" in line 17 is closest in meaning to
(A) instrument (B) variety (C) surplus (D) symbol
20. It can be inferred that research in other academic fields relates to research in the author's field in which of the following ways?
(A) It contributes relevant information
(B) It is carried out on a simpler level.
(C) It is less reliable than research in the author's field.
(D) It causes misunderstandings if applied to the author's field.
21. Which of the following conclusions is supported by the passage?
(A) Counting processes did not develop until after writing became widespread.
(B) Early counting methods required herds of animals.
(C) Mathematics has remained unchanged since ancient times.
(D) Early humans first counted because of necessity.
22. Where in the passage does the author mention the ability of animals to recognized small and large groups?
(A) Lines 1-2 (B) Lines 6-9 (C) Lines 10-12 (D) Lines 17-18

Questions 23-31

As the merchant class expanded in the eighteenth-century North American colonies, the silversmith and the coppersmith businesses rose to serve it. Only a few silversmiths were available in New York or Boston in the late seventeenth century, but in the eighteenth century they could be found in all major colonial cities. No other colonial artisans rivaled the silversmiths' prestige. They handled the most expensive materials and possessed direct connections to prosperous colonies merchants. Their products, primarily silver plates and bowls, reflected their exalted status and testified to their customers' prominence.

Silver stood as one of the surest ways to store wealth at a time before neighborhood banks existed. Unlike the silver coins from which they were made, silver articles were readily identifiable. Often formed to individual specifications, they always carried the silversmith's distinctive markings and consequently could be traced and retrieved. Customers generally secured the silver for the silver objects they ordered. They saved coins, took them to smiths, and discussed the type of pieces they desired. Silversmiths complied with these requests by melting the money in a small furnace, adding a bit of copper to form a stronger alloy, and casting the alloy in rectangular blocks. They hammered these ingots to the appropriate thickness by hand, shaped them, and pressed designs into them for adornment. Engraving was also done by hand. In addition to plates and bowls, some customers sought more intricate products, such as silver teapots. These were made by shaping or casting parts separately and then soldering them together.

Colonial coppersmithing also came of age in the early eighteenth century and prospered in northern cities. Copper's ability to conduct heat efficiently and to resist corrosion contributed to its attractiveness. But because it was expensive in colonial America, coppersmiths were never very numerous. Virtually all copper worked by smiths was imported as sheets or obtained by recycling old copper goods. Copper was used for practical items, but it was not admired for its beauty. Coppersmiths employed it to fashion pots and kettles for the home. They shaped it in much the same manner as silver or melted it in a foundry with lead or tin. They also mixed it with zinc to make brass for maritime and scientific instruments.

23. According to the passage, which of the following eighteenth-century developments had a strong impact on silversmiths?
- (A) a decrease in the cost of silver
 - (B) the invention of heat-efficient furnaces
 - (C) the growing economic prosperity of colonial merchants
 - (D) the development of new tools used to shape silver
24. The word "They" in line 5 refers to
- (A) silversmiths
 - (B) major colonial cities
 - (C) other colonial artisans
 - (D) materials
25. The word "exalted" in line 7 is closest in meaning to
- (A) unusual
 - (B) uncertain
 - (C) surprising
 - (D) superior
26. In colonial America, where did silversmiths usually obtain the material to make silver articles?
- (A) From their own mines
 - (B) From importers
 - (C) From other silversmiths
 - (D) From customers
27. The word "ingots" in line 17 refers to
- (A) coins that people saved
 - (B) blocks of silver mixed with copper
 - (C) tools used to shape silver plates
 - (D) casts in which to form parts of silver articles
28. The phrase "came of age" in line 22 is closest in meaning to
- (A) established itself
 - (B) declined
 - (C) became less expensive
 - (D) was studied
29. The passage mentions all of the following as uses for copper in colonial America EXCEPT
- (A) cooking pots
 - (B) scientific instruments
 - (C) musical instruments
 - (D) maritime instruments
30. According to the passage, silversmiths and coppersmiths in colonial America were similar in which of the following ways?
- (A) The amount of social prestige they had
 - (B) The way they shaped the metal they worked with
 - (C) The cost of the goods they made
 - (D) The practicality of goods they made
31. Based on the information in paragraph 4, which of the following was probably true about copper in the colonies?
- (A) The copper used by colonists was not effective in conducting heat.
 - (B) The copper items created by colonial coppersmiths were not skillfully made.
 - (C) There were no local copper mines from which copper could be obtained.
 - (D) The price of copper suddenly decreased.

Questions 32-40

Fossils are the remains and traces (such as footprints or other marks) of ancient plant and animal life that are more than 10,000 years old. They range in size from microscopic structures to dinosaur skeletons and complete bodies of enormous animals.

Line Skeletons of extinct species of human are also considered fossils.

- (5) An environment favorable to the growth and later preservation of organisms is required for the occurrence of fossils. Two conditions are almost always present:
(1) The possession of hard parts, either internal or external, such as bones, teeth, scales, shells, and wood; these parts remain after the rest of the organism has decayed. Organisms that lack hard parts, such as worms and jelly fish, have left a meager
(10) geologic record. (2) Quick burial of the dead organism, so that protection is afforded against weathering, bacterial action, and scavengers.

Nature provides many situations in which the remains of animals and plants are protected against destruction. Of these, marine sediment is by far the most important environment for the preservation of fossils, owing to the incredible richness of marine
(15) life. The beds of former lakes are also prolific sources of fossils. The rapidly accumulating sediments in the channels, floodplains, and deltas of streams bury fresh-water organisms, along with land plants and animals that fall into the water. The beautifully preserved fossil fish from the Green River soil shale of Wyoming in the western United States lived in a vast shallow lake.

- (20) The frigid ground in the far north acts as a remarkable preservative for animal fossils. The woolly mammoth, along-haired rhinoceros, and other mammals have been periodically exposed in the tundra of Siberia, the hair and red flesh still frozen in cold storage.

Volcanoes often provide environments favorable to fossil preservation. Extensive

- (25) falls of volcanic ash and coarser particles overwhelm and bury all forms of life, from flying insects to great trees.

Caves have preserved the bones of many animals that died in them and were subsequently buried under a blanket of clay or a cover of dripstone. Predatory animals and early humans alike sought shelter in caves and brought food to them to the eater,

- (30) leaving bones that paleontologists have discovered.

32. The passage primarily discusses which of the following?

- (A) Types of fossils found in different climates
- (B) What is learned from studying fossils
- (C) Conditions favorable to the preservation of fossils
- (D) How fossils are discovered

33. The word "traces" in line 1 is closest in meaning to

- (A) structures
- (B) importance
- (C) skeletons
- (D) imprints

34. All of the following facts about fossils are referred to by the author (paragraph 1) EXCEPT the fact that they can be

- (A) microscopically small
- (B) skeletons of human ancestors
- (C) complete animal bodies
- (D) fragile

35. The fossil fish from the Green River (paragraph 3) were probably preserved because they were
(A) in a deep lake (B) covered by sediment
(C) protected by oil (D) buried slowly
36. The word "exposed" in line 22 is closest in meaning to
(A) photographed (B) uncovered (C) located (D) preserved
37. Which of the following is LEAST likely to be found as a fossil, assuming that all are buried rapidly?
(A) a dinosaur (B) a woolly mammoth
(C) a human ancestor (D) a worm
38. It can be inferred that a condition that favors fossilization when volcanic ash falls to Earth is
(A) quick burial (B) cold storage (C) high temperature (D) lack of water
39. The word "them" in line 29 refers to
(A) predatory animals (B) early humans (C) caves (D) bones
40. Which of the following is true of the environments in which fossil are found?
(A) Very different environments can favor fossilization.
(B) There are few environments in which fossils are protected.
(C) Environments that favor fossilization have similar climates.
(D) Environments that favor fossilization support large populations of animals.

Questions 41-50

- A useful definition of an air pollutant is a compound added directly or indirectly by humans to the atmosphere in such quantities as to affect humans, animals vegetations, or materials adversely. Air pollution requires a very flexible definition that permits continuous change. When the first air pollution laws were established in England in the fourteenth century, air pollutants were limited to compounds that could be seen or smelled—a far cry from the extensive list of harmful substances known today. As technology has developed and knowledge of the health aspects of various chemicals has increased, the list of air pollutants has lengthened. In the future, even water vapor might be considered an air pollutant under certain conditions.
- (10) Many of the more important air pollutants, such as sulfur oxides, carbon monoxide, and nitrogen oxides, are found in nature. As the Earth developed, the concentrations of these pollutants were altered by various chemical reactions; they became components in biogeochemical cycle. These serve as an air purification scheme by allowing the compounds to move from the air to the water or soil on a global basis, nature's
- (15) output of these compounds dwarfs that resulting from human activities. However, human production usually occurs in a localized area, such as a city.

- In this localized regions, human output may be dominant and may temporarily overload the natural purification scheme of the cycle. The result is an increased concentration of noxious chemicals in the air. The concentrations at which the adverse effects appear will be greater than the concentrations that the pollutants would have in the absence of human activities. The actual concentration need not be large for a substance to be a pollutant; in fact the numerical value tells us little until we know how much of an increase this represents over the concentration that would occur naturally in the area. For example, sulfur dioxide has detectable health effects at
- (20) 0.08 parts per million (ppm), which is about 400 times its natural level. Carbon monoxide, however, has a natural level of 0.1 ppm and is not usually a pollutant until its level reaches about 15 ppm.

41. What does the passage mainly discuss?
- (A) The economic impact of air pollution
(B) What constitutes an air pollutant
(C) How much harm air pollutants can cause
(D) The effects of compounds added to the atmosphere
42. The word "adversely" in line 3 is closest in meaning to
- (A) negatively (B) quickly (C) admittedly (D) considerably
43. It can be inferred from the first paragraph that
- (A) water vapor is an air pollutant in localized areas
(B) most air pollutants today can be seen or smelled
(C) the definition of air pollution will continue to change
(D) a substance becomes an air pollutant only in cities
44. The word "altered" in line 12 is closest in meaning to
- (A) eliminated (B) caused (C) slowed (D) changed

45. Natural pollutants can play an important role in controlling air pollution for which of the following reasons?
- (A) They function as part of a purification process.
 - (B) They occur in greater quantities than other pollutants.
 - (C) They are less harmful to living beings than are other pollutants.
 - (D) They have existed since the Earth developed.
46. According to the passage, which of the following is true about human-generated air pollution in localized regions?
- (A) It can be dwarfed by nature's output of pollutants in the localized region.
 - (B) It can overwhelm the natural system that removes pollutants.
 - (C) It will damage areas outside of the localized regions.
 - (D) It will react harmfully with naturally occurring pollutants.
47. The word "noxious" in line 19 is closest in meaning to
- (A) harmful
 - (B) noticeable
 - (C) extensive
 - (D) weak
48. According to the passage, the numerical value of the concentration level of a substance is only useful if
- (A) the other substances in the area are known
 - (B) it is in a localized area
 - (C) the naturally occurring level is also known
 - (D) it can be calculated quickly
49. The word "detectable" in line 24 is closest in meaning to
- (A) beneficial
 - (B) special
 - (C) measurable
 - (D) separable
50. Which of the following is best supported by the passage?
- (A) To effectively control pollution local government should regularly review their air pollution laws.
 - (B) One of the most important steps in preserving natural lands is to better enforce air pollution laws.
 - (C) Scientists should be consulted in order to establish uniform limits for all air pollutants.
 - (D) Human activities have been effective in reducing air pollution.